

How Small is Small?

Non-linearities in Heterogeneous Agent Models

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Workshop on Methods and Applications for Dynamic Equilibrium Models

OVERVIEW

- Heterogeneous-agent models are a workhorse framework for policy analysis
 - ▶ Match the level and distribution of MPCs in the data

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- BUT: How small is small?
- Our answer: for fiscal transfers, “small” must be *very small*

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- In calibrated HA models, MPCs are **highly non-linear** in wealth
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ROADMAP

1. Partial equilibrium environment
2. Fiscal stimulus in general equilibrium
3. Other shocks and results

HOUSEHOLD CONSUMPTION-SAVING PROBLEM

- Continuum of households with idiosyncratic income shocks

$$\max_{\{c_{i,t}, a_{i,t}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_{i,t})$$

$$\text{s.t.} \quad c_{i,t} + a_{i,t} = (1+r)a_{i,t-1} + (1-\tau)y_{i,t} + T_t \quad \text{and} \quad a_{i,t} \geq 0$$

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Intertemporal Euler equation:

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Optimization + aggregation $\Rightarrow$ aggregate consumption function:

$$\mathcal{C}_t(\{T_s\})$$

FISCAL TRANSFER POLICY

Starting from steady state, a one-time unanticipated lump-sum transfer at $t = 0$ [MIT-shock]

$$T_0 = T + \Delta, \quad T_t = T \text{ for } t > 0$$

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Object of interest. Aggregate consumption response:

$$C_t(\Delta) - C_t(0)$$

MODEL COMPUTATION: NON-LINEAR VS. LINEAR TRANSITIONS

Steady-state $\mathcal{C}_0$: endogenous-grid point method for decision rules + aggregation

MODEL COMPUTATION: NON-LINEAR VS. LINEAR TRANSITIONS

Non-linear

- Use household policy to solve exactly consumption response

$$\mathcal{C}_0(\Delta) = \int_Y \int_{-\infty}^{\infty} c(\tilde{x} + \Delta, y) \, dG(\tilde{x} \mid y) \, dF(y)$$

cash-on-hand before transfer

Linear

- Taylor expansion of aggregate consumption around steady state

$$\mathcal{C}_0(\Delta) \approx \mathcal{C}_0(0) + \mathcal{C}'_0(0) \cdot \Delta$$

Slopes computed by finite differences or SSJ toolbox. Then, extrapolate.

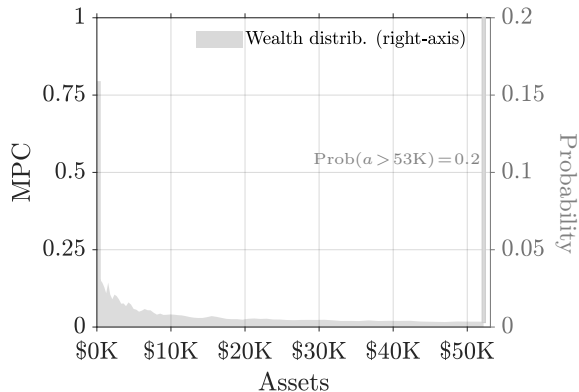
CALIBRATION

- A period = a quarter. $u(c) = \log(c)$
- Earnings process, $y_{i,t}$, calibrated to match second moments of annual earnings (Kaplan and Violante 2022)
- Interest rate 2% annual
- Tax rate $(\tau) = 27.4\%$
- Steady-state transfers = 15% of GDP ($\approx \$25\text{K}/\text{yr}$)
- Heterogeneous discount factors:

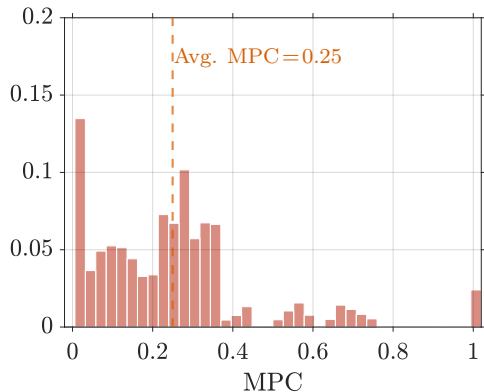
$$\left. \begin{array}{l} \text{Mean wealth / annual GDP} = 100\% (\approx \$166\text{K}) \\ \text{Avg. quarterly MPC (out of \$1,000)} = 25\% \end{array} \right\} \Rightarrow \bar{\beta} = 0.972, \sigma_{\beta} = 0.010$$

STEADY-STATE WEALTH DISTRIBUTION AND MPCs

(a) Wealth distribution

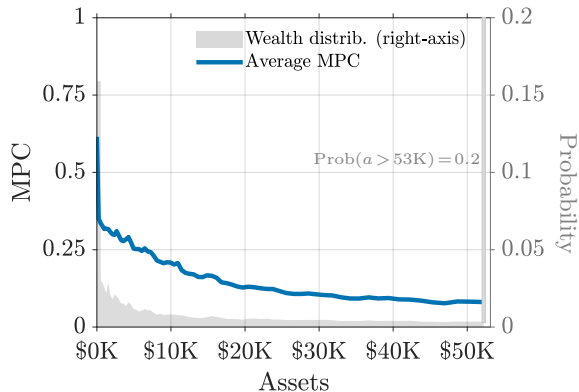


(b) MPC distribution

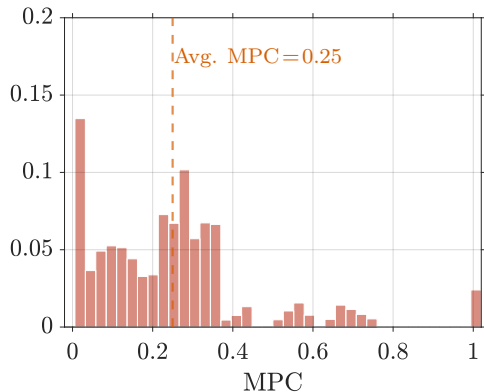


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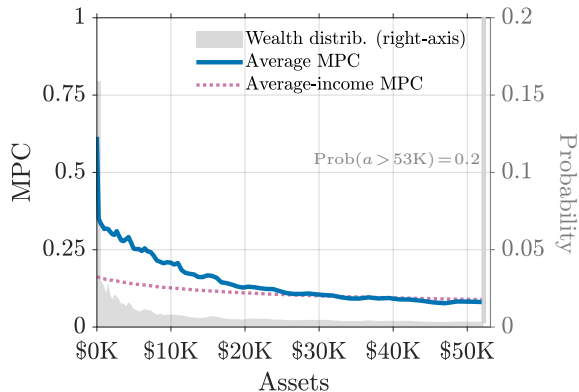


(b) MPC distribution

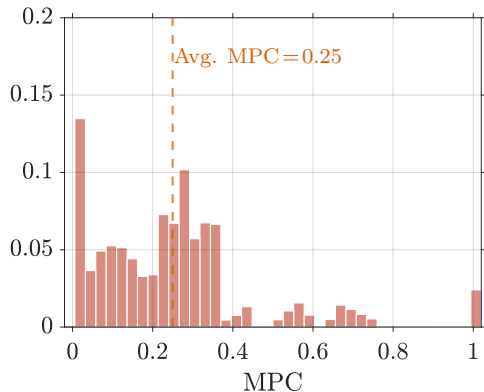


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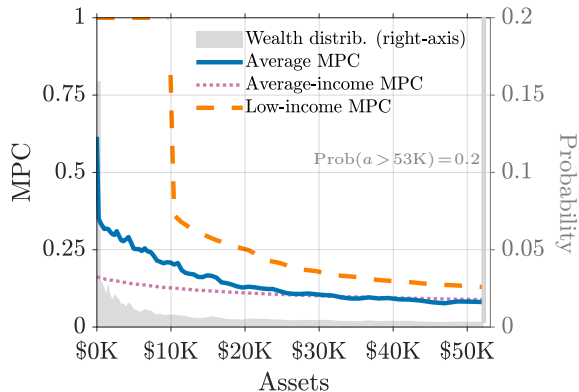


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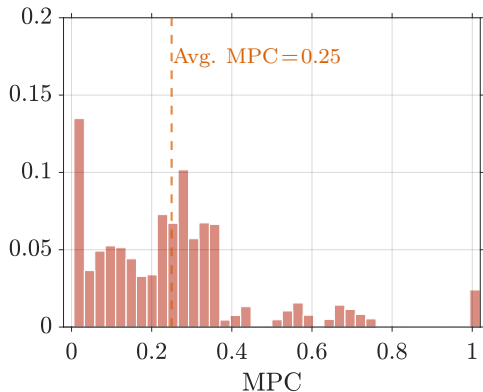


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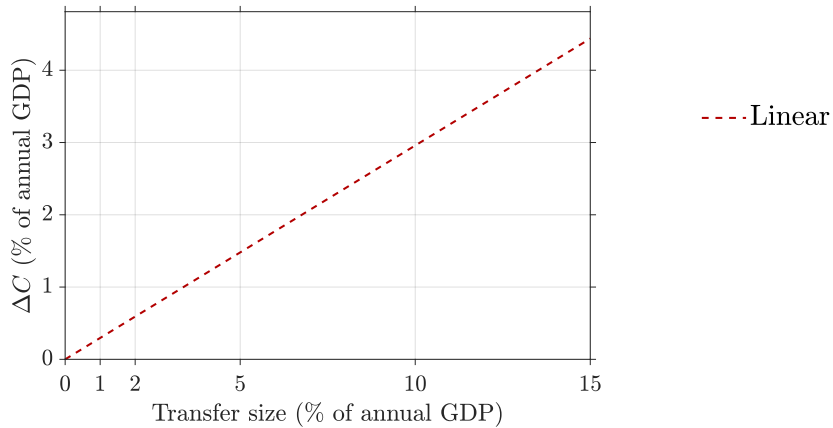
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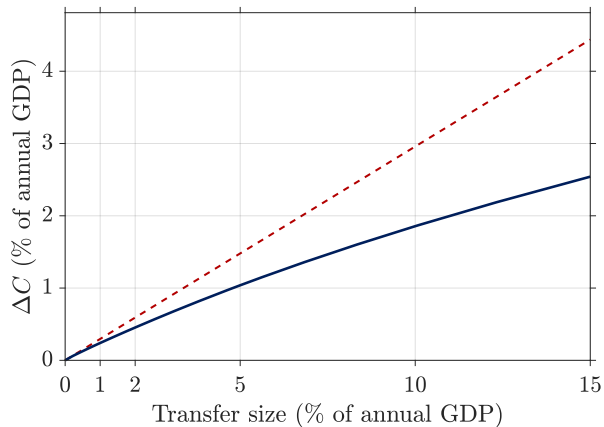
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AGGREGATE CONSUMPTION RESPONSE



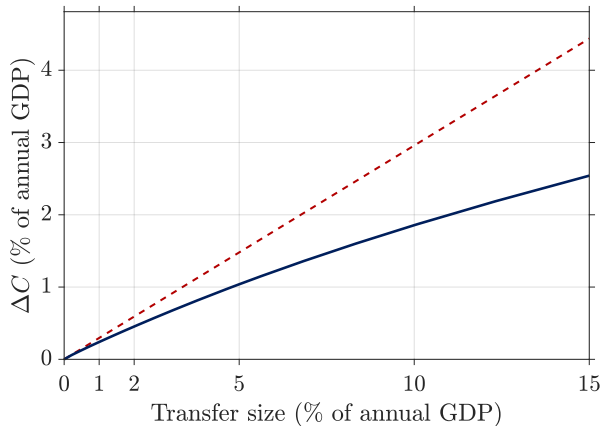
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--- Linear
— Non-linear

- Non-linear solution is concave

AGGREGATE CONSUMPTION RESPONSE



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- Non-linear solution is concave
- Linear overstates response:
 - ▶ At $\Delta = 5\%$ of GDP ($\approx \$8K$):
overstates by 42%
 - ▶ At $\Delta = 0.5\%$: overstates by 17%

UNDERSTANDING THE GAP: TWO SOURCES OF CONCAVITY

- Concavity of the individual consumption function
- Households moving on and off the borrowing constraint

UNDERSTANDING THE GAP: TWO SOURCES OF CONCAVITY

$$\mathcal{C}_0(\Delta) = \int_Y \int_{-\infty}^{\infty} c(\tilde{x} + \Delta, y) \, dG(\tilde{x} \mid y) \, dF(y)$$

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Initial distribution (before transfers)

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Initial distribution (before transfers)

- Denote $\bar{x}(y)$ cash-on-hand at which the borrowing constraint stops binding
 - ▶ Household constrained if $\tilde{x} + \Delta \leq \bar{x}(y)$

UNDERSTANDING THE GAP: TWO SOURCES OF CONCAVITY

$$\mathcal{C}_0(\Delta) = \int_Y \left[\underbrace{\int_{-\infty}^{\bar{x}(y)-\Delta} c(\tilde{x} + \Delta, y) \, dG(\tilde{x} \mid y)}_{\text{constrained}} + \underbrace{\int_{\bar{x}(y)-\Delta}^{\infty} c(\tilde{x} + \Delta, y) \, dG(\tilde{x} \mid y)}_{\text{unconstrained}} \right] dF(y)$$

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MPC has a discontinuity at $\bar{x}(y)$.

$$c_x(x, y) = \begin{cases} 1, & x < \bar{x}(y) \quad (\text{constrained}), \\ c_x^+(x, y) < 1, & x > \bar{x}(y) \quad (\text{unconstrained}). \end{cases}$$

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$$\mathcal{C}'_0(0) = \bar{\Lambda} \times \underbrace{1}_{\text{MPC}=1} + (1 - \bar{\Lambda}) \times \underbrace{\mathbb{E}[c_x^+ | \tilde{x} > \bar{x}(y)]}_{\text{unconstrained MPC}}$$

where $\Lambda(\Delta)$ is the fraction of constrained households and $\bar{\Lambda} = \Lambda(0)$



$$\Lambda(\Delta) \equiv \int_Y G(\bar{x}(y) - \Delta | y) dF(y)$$

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Linear solution simply extrapolates steady-state aggregate MPC

- Changes in the constrained share are irrelevant to first order
 - As if constrained households maintain $\text{MPC} = 1$ regardless of Δ

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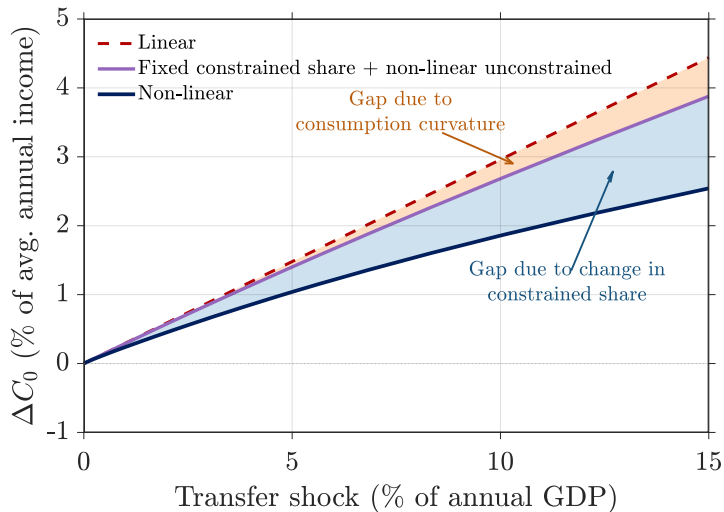
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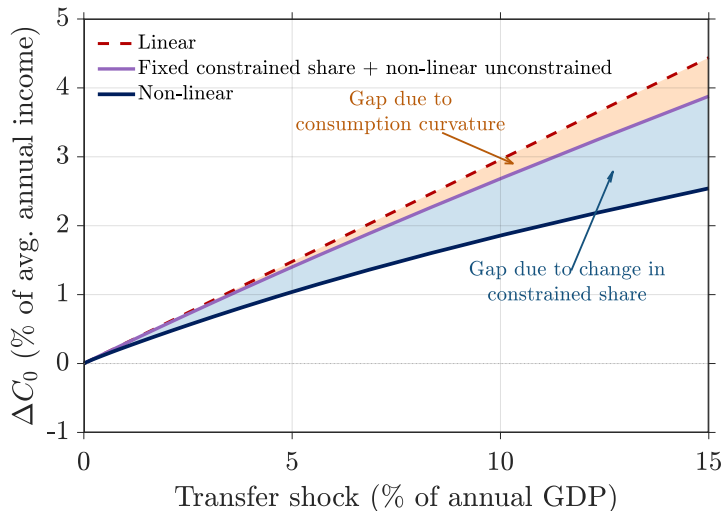
change in constrained share

$C''_0(0) \Rightarrow$ The aggregate consumption response is **concave** in transfer size.

DECOMPOSING THE LINEAR-NON-LINEAR GAP



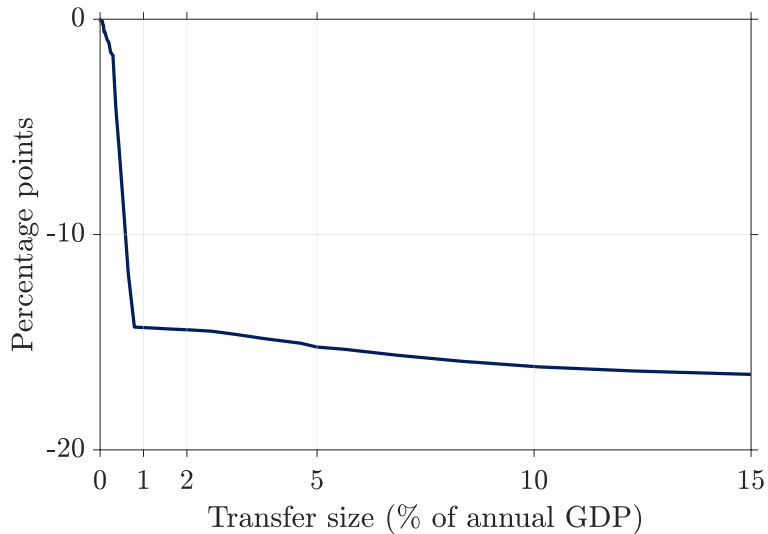
DECOMPOSING THE LINEAR-NON-LINEAR GAP



1% transfer
Constrained share 93%
Curvature 7%

5% transfer
Constrained share 82%
Curvature 18%

CHANGE IN SHARE OF CONSTRAINED HOUSEHOLDS

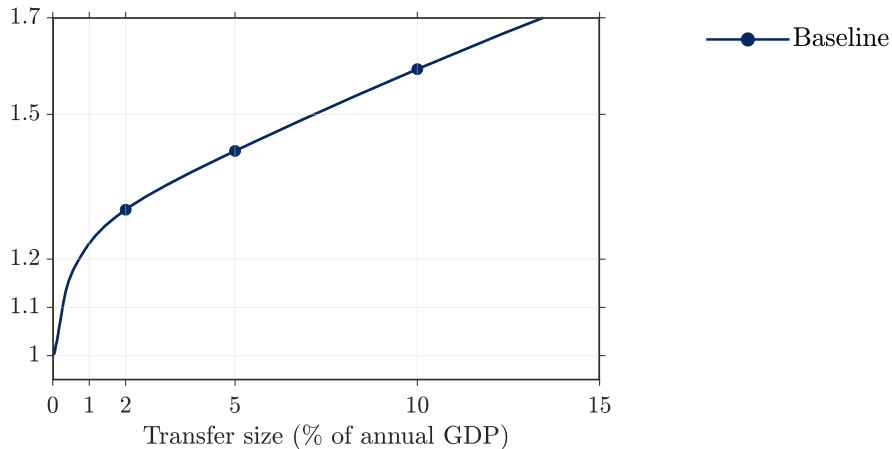


TAKING STOCK: WHY LINEAR-METHOD OVERSTATES?

- Linear solution, changes in constrained share have zero first-order effects
 - ▶ Effectively, as if constrained households permanently $MPC=1$
 - Non-linear solution is highly concave
 - ▶ Key: Drop in MPC at kink $\times$ Share of households becoming unconstrained
- $\Rightarrow$ Significant gaps between linear and non-linear solution, even for small shocks

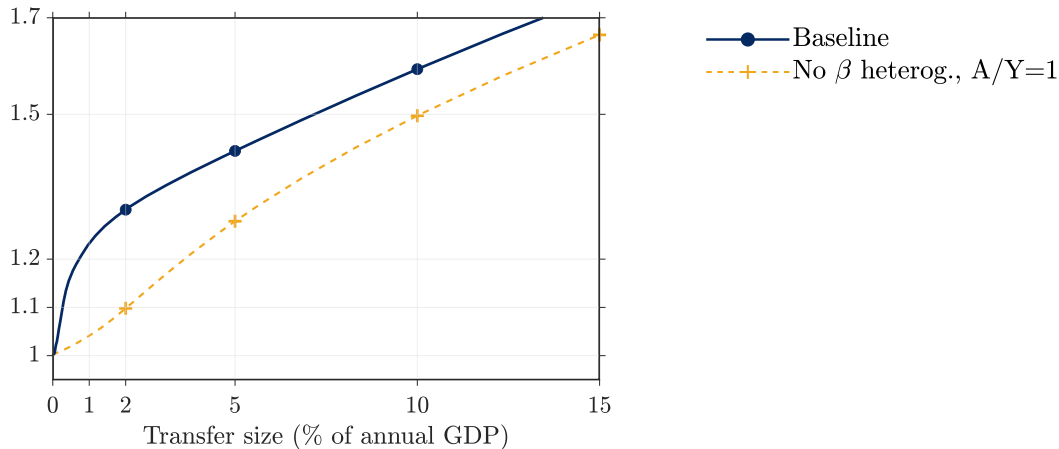
HOW SMALL IS SMALL? ROBUSTNESS OF P.E. RESULTS

Linear/Non-linear Consumption Response



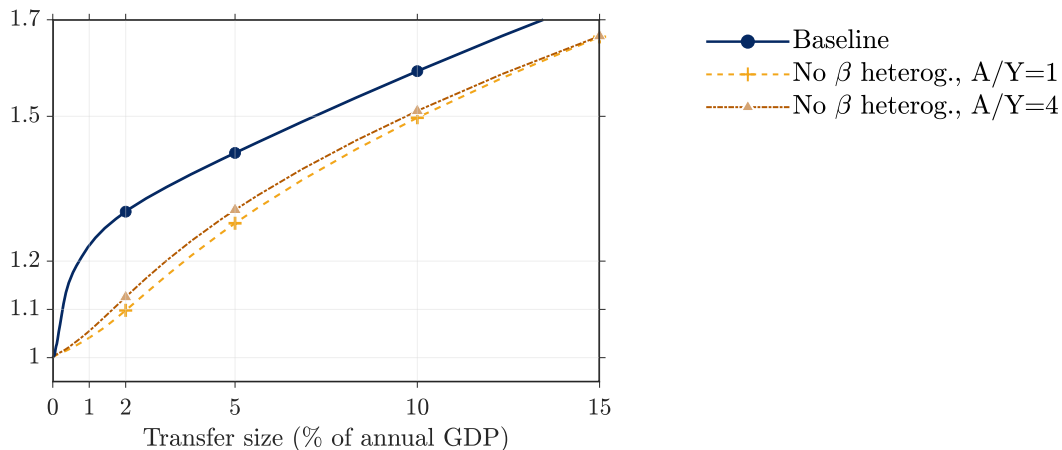
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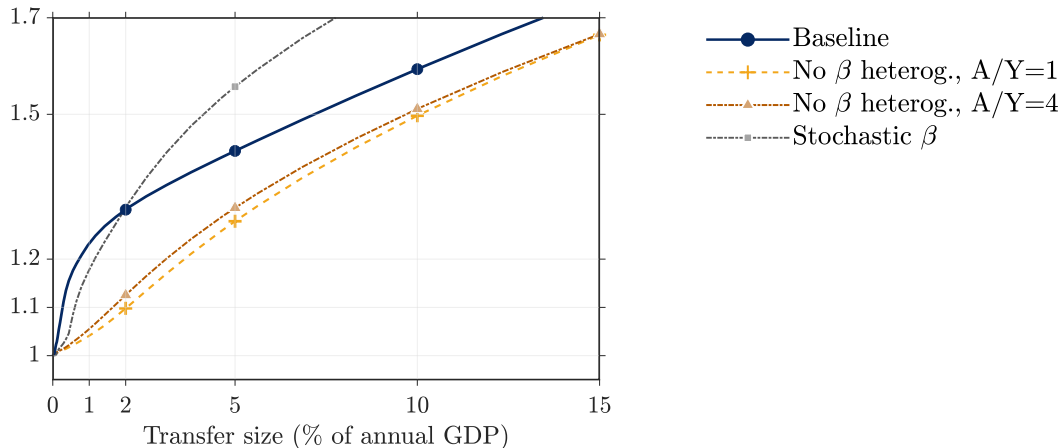
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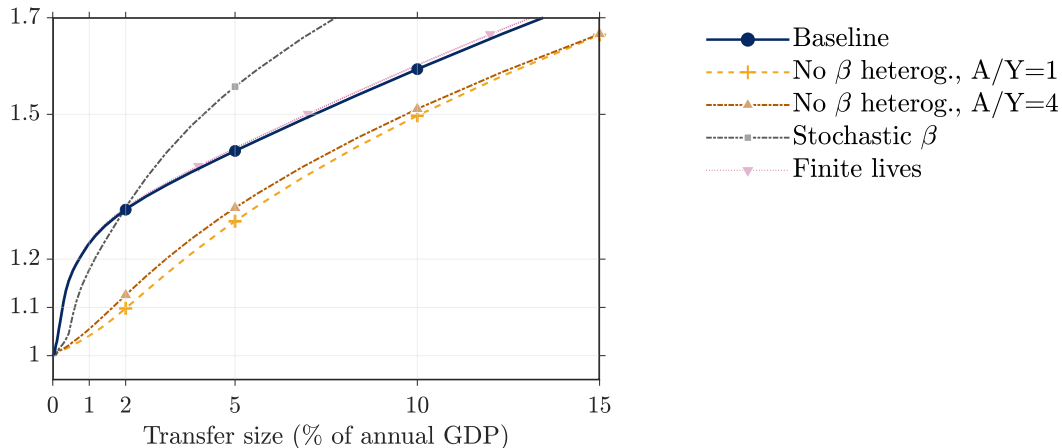
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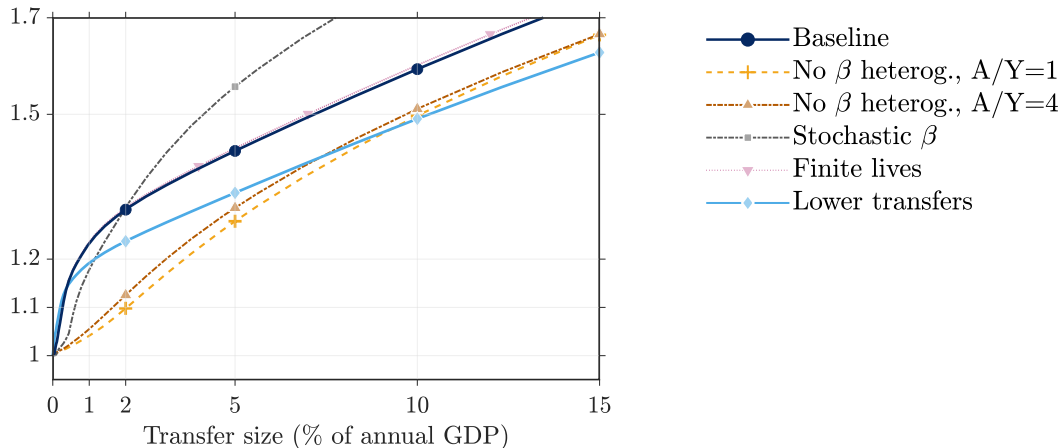
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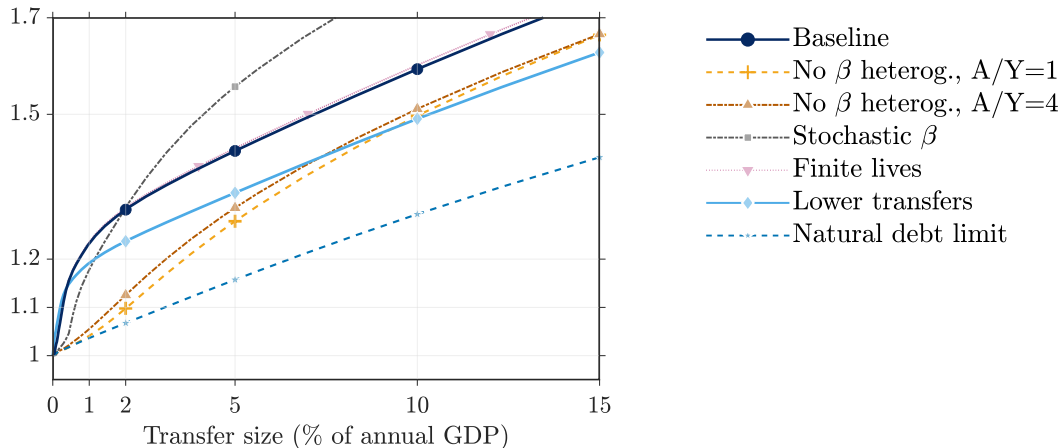
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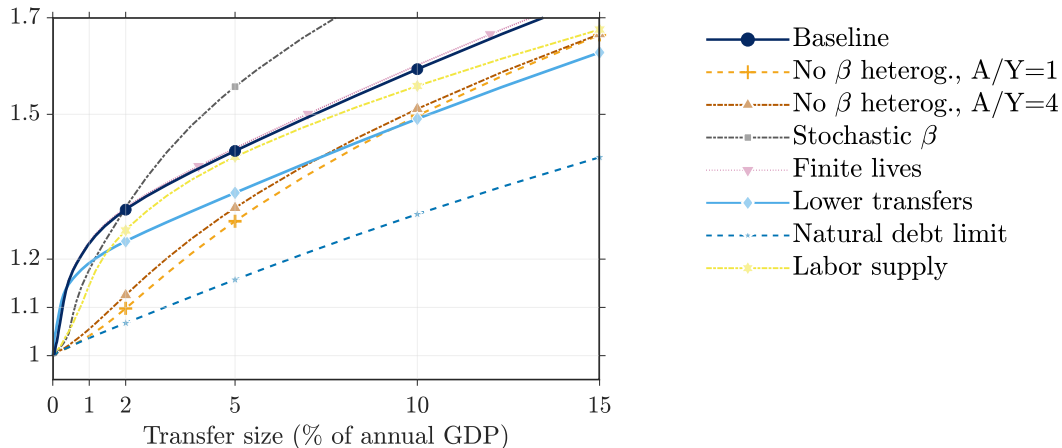
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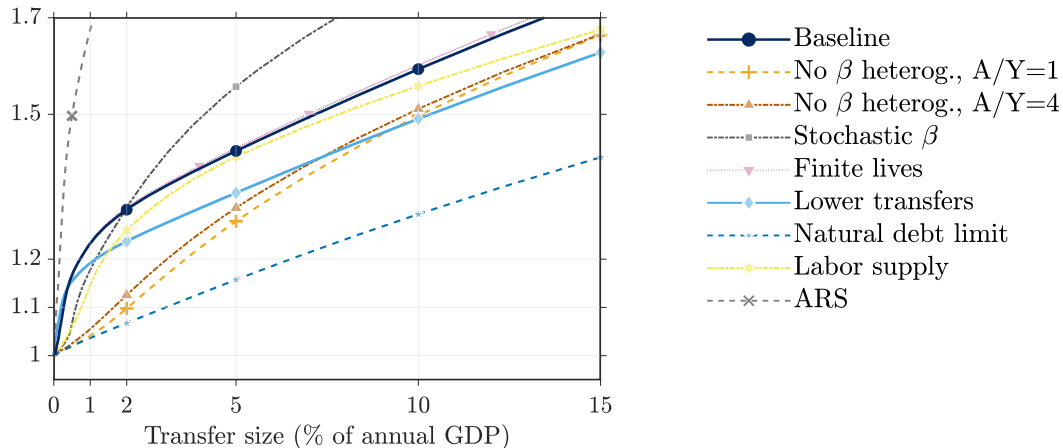
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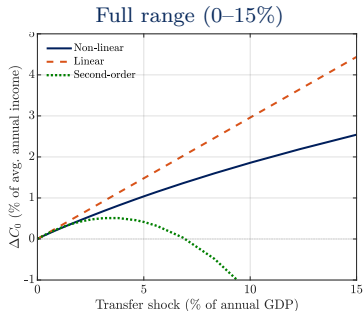
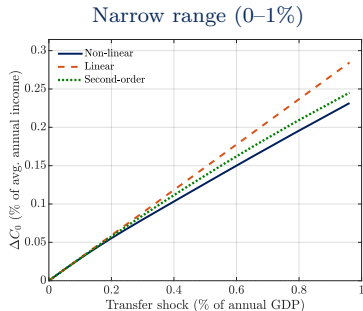


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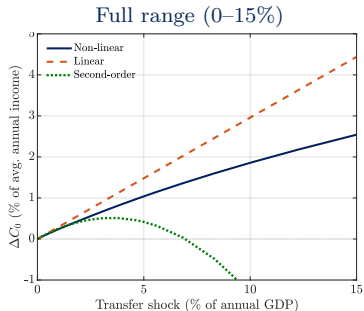
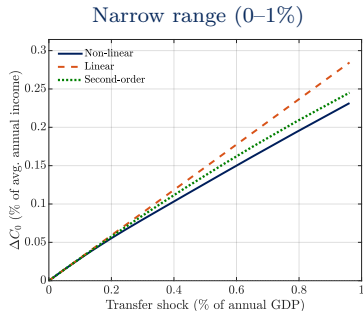


DO HIGHER-ORDER LOCAL METHODS HELP?



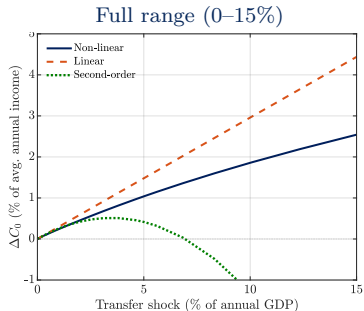
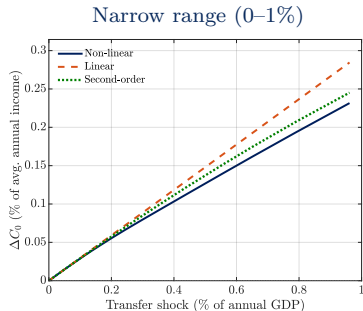
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DO HIGHER-ORDER LOCAL METHODS HELP?



- Second-order improves for narrow shocks, but deteriorates quickly

DO HIGHER-ORDER LOCAL METHODS HELP?



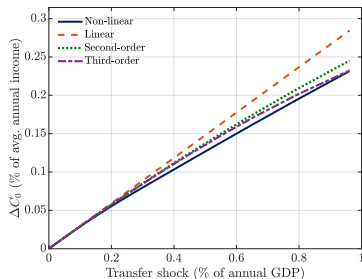
$$C_0''(0) = \underbrace{\Lambda'(0) \mathbb{E}[(1 - c_x^+) \mid \tilde{x} = \bar{x}(y)]}_{\text{movement off constraint} < 0} + \underbrace{(1 - \bar{\Lambda}) \mathbb{E}[c_{xx}^+ \mid \tilde{x} > \bar{x}(y)]}_{\text{individual curvature} < 0}$$

$$\Lambda'(0) = - \int_Y g(\bar{x}(y) \mid y) dF(y) < 0$$

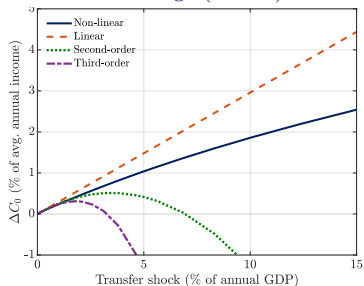
Second-order extrapolates change in constrained share at constant rate $\Rightarrow$ overshoots for small Δ , *understating* aggregate response

DO HIGHER-ORDER LOCAL METHODS HELP?

Narrow range (0–1%)



Full range (0–15%)



$$C_0'''(0) = \underbrace{\Lambda''(0)}_{\text{curvature of constrained share}} \tilde{\mathbb{E}}[(1 - c_x^+) | \tilde{x} = \bar{x}(y)] - \Lambda'(0) \mathbb{E}[c_{xx}^+ | \tilde{x} = \bar{x}(y)] + (1 - \bar{\Lambda}) \mathbb{E}[c_{xxx}^+ | \tilde{x} > \bar{x}(y)]$$

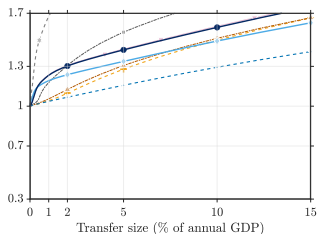
curvature of constrained share

$$\Lambda''(0) = \int_Y g'(\bar{x}(y) | y) dF(y) > 0$$

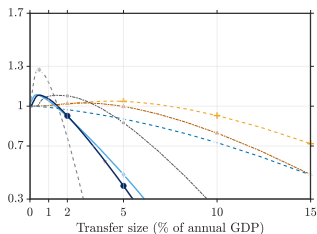
Third-order overshoots *further*

DO HIGHER-ORDER APPROXIMATIONS HELP?

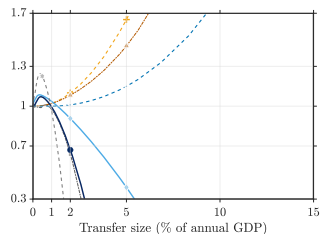
Linear / non-linear



Second order / non-linear

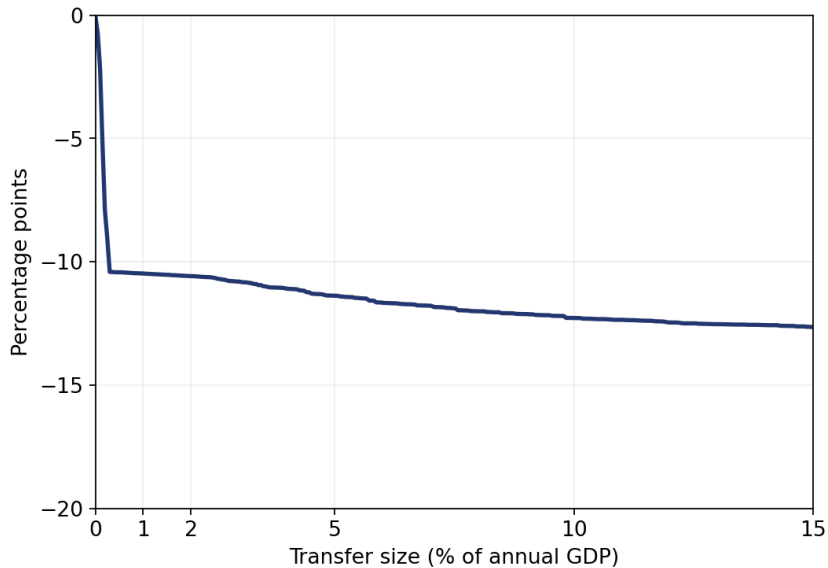


Third order / non-linear

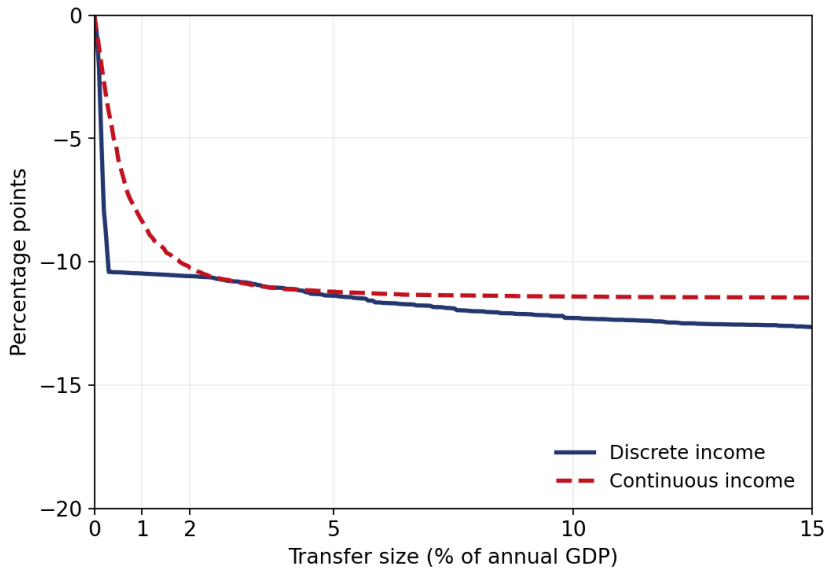


—●— Baseline - - + - - No β heterog., $A/Y=1$ - - ▲ - - No β heterog., $A/Y=4$ - - ■ - - Stochastic β
 - - ▼ - - Finite lives —◆— Lower transfers - - * - - Natural debt limit - - * - - ARS

CHANGE IN SHARE OF CONSTRAINED HOUSEHOLDS

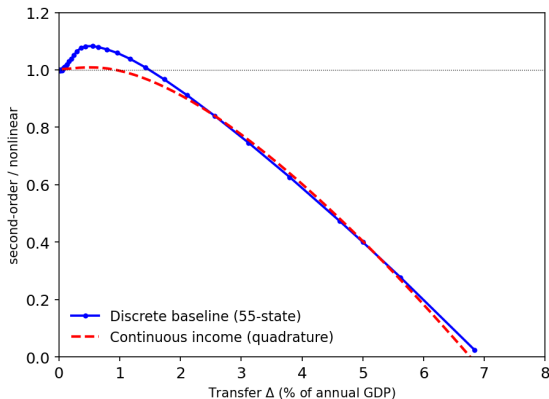


CHANGE IN SHARE OF CONSTRAINED HOUSEHOLDS



ROBUSTNESS TO DISCRETENESS OF INCOME PROCESS

Second-order/Non-Linear



A continuous income process yields a smoother higher-order approximation, but the approximation remains accurate only for small shocks.

ROADMAP

1. Partial equilibrium model and calibration
2. Fiscal stimulus in general equilibrium
3. Other shocks

GENERAL EQUILIBRIUM MODEL

PE + production, labor supply, nominal rigidities, fiscal/monetary rules.

GENERAL EQUILIBRIUM MODEL

- **Households:**

- ▶ Utility flow, $u(c_{it}) - v(n_{it})$ with $v' > 0, v'' > 0$
- ▶ Income $(1 - \tau) \left(\frac{W_t}{P_t} e_{it} n_{it} + d_{i,t} \right)$

GENERAL EQUILIBRIUM MODEL

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- ▶ Labor union with sticky wages (linearized)

$$\pi_t = \kappa \left(v'(N_t) - \frac{1 - \tau}{C_t} \right) + \bar{\beta} \pi_{t+1}, \quad (\text{Phillips Curve})$$

GENERAL EQUILIBRIUM MODEL

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- **Firms and Technology:**

$$\left. \begin{array}{l} \text{Production: } Y_t = N_t \\ \text{Flexible prices} \end{array} \right\} \Rightarrow W_t = P_t, \quad d_{i,t} = 0$$

GENERAL EQUILIBRIUM MODEL (CTD)

- Government budget:

$$B_t + \tau Y_t = G + T_t + (1 + r_{t-1})B_{t-1}$$

- Fiscal rule ($t > 0$)

$$T_t = T - \phi_B \bar{r}(B_{t-1} - \bar{B})$$

- Monetary policy:

$$i_t = \bar{i} + \phi_\pi(\pi_t - \bar{\pi})$$

CALIBRATION

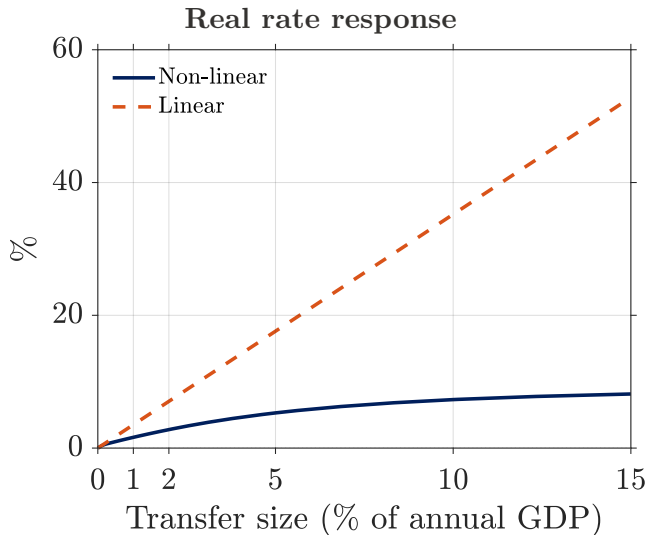
Additional Parameters for GE Model

Parameter	Value	Description
$\bar{B}$	4	Steady-state government debt=100% GDP
κ	0.01	Phillips curve slope
ν	1	Frisch elasticity of labor supply
ϕ_π	1.5	Taylor coefficient
ϕ_B	8	Fiscal rule
G	0.104	Government spending
T	0.15	Government transfers

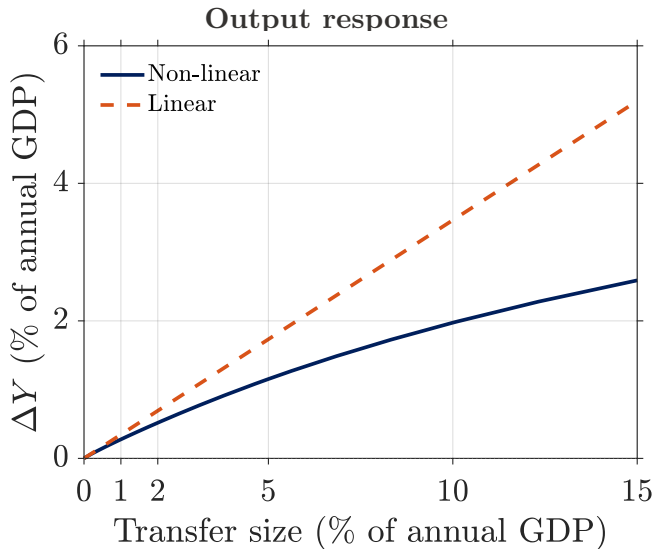
Computation:

- Linear: sequence space Jacobian (ABRS)
- Non-linear: quasi-Newton algorithm

FLEXIBLE-PRICE ALLOCATION



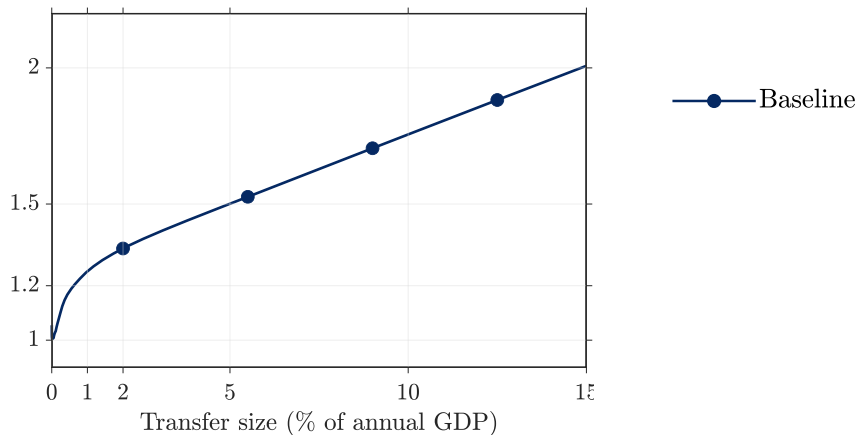
- No output effects
- Linear methods overstate demand increase and rise in r



- At $\Delta = 2\%$ GDP: linear overstates by 34%
- At $\Delta = 5\%$ GDP: linear overstates by 50%

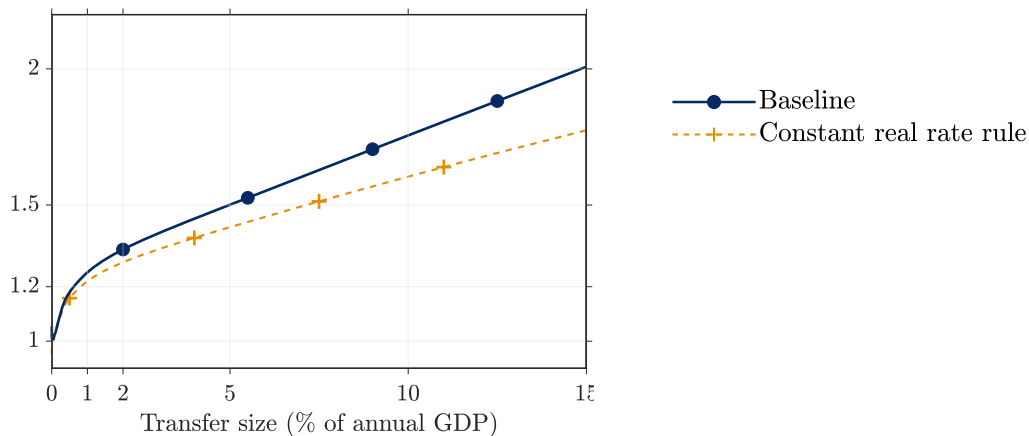
HOW SMALL IS SMALL? ROBUSTNESS IN HANK

Linear/Non-linear Output Response



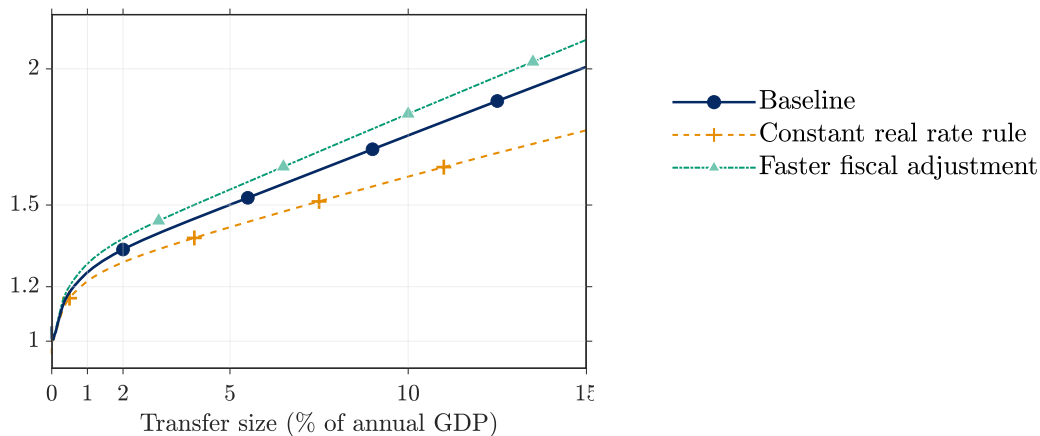
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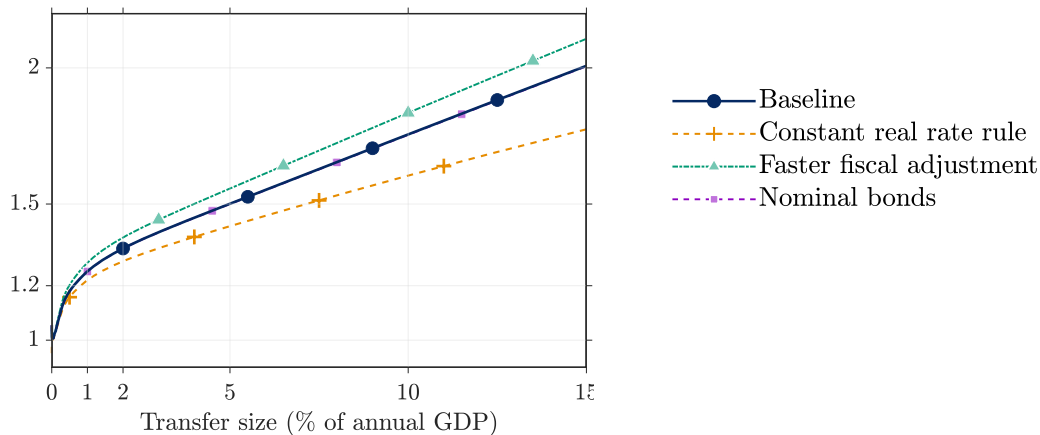
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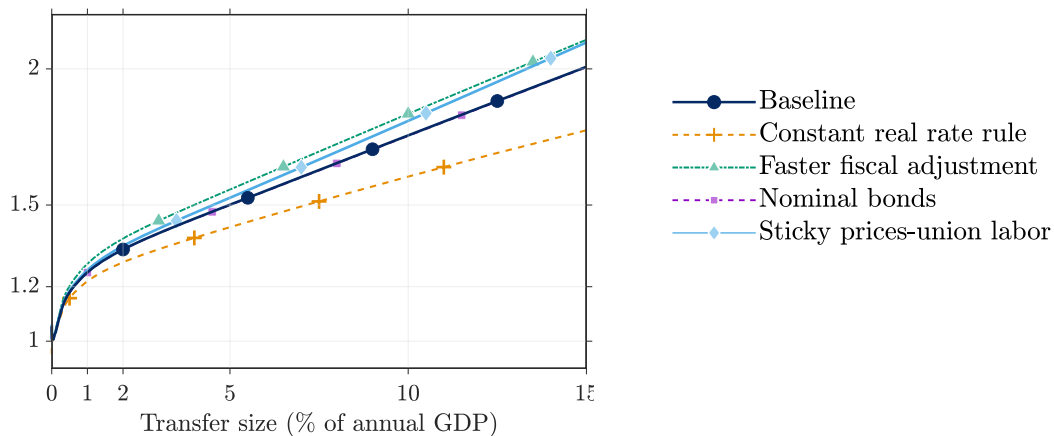
HOW SMALL IS SMALL? ROBUSTNESS IN HANK

Linear/Non-linear Output Response



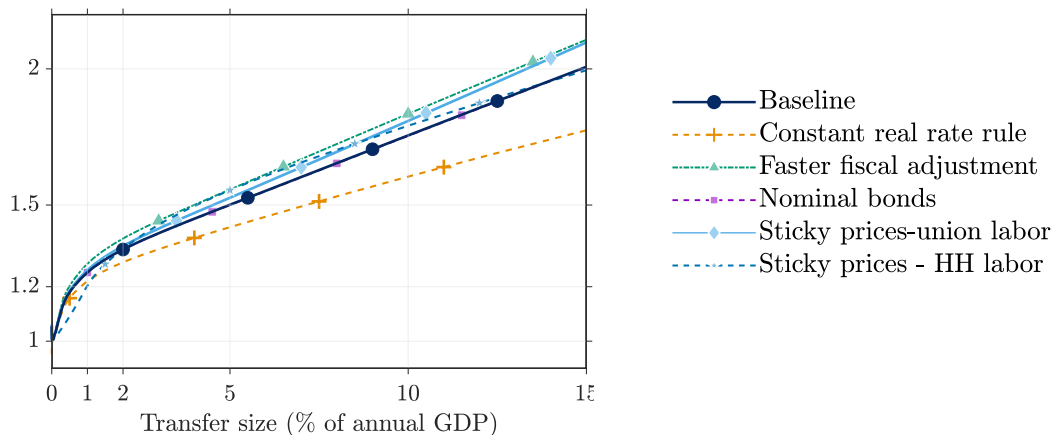
HOW SMALL IS SMALL? ROBUSTNESS IN HANK

Linear/Non-linear Output Response



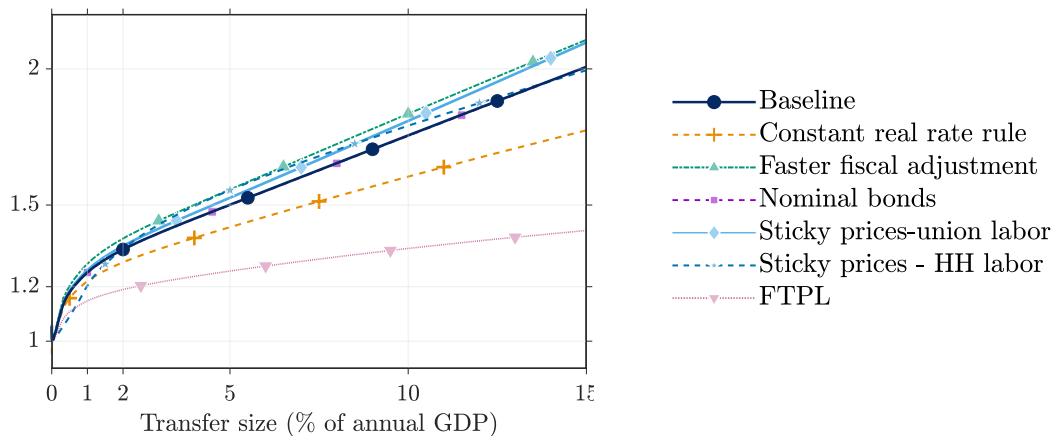
HOW SMALL IS SMALL? ROBUSTNESS IN HANK

Linear/Non-linear Output Response



HOW SMALL IS SMALL? ROBUSTNESS IN HANK

Linear/Non-linear Output Response



OTHER RESULTS

1. Fiscal contractions vs. expansions

go →

2. Gap in persistence

go →

3. Targeted transfers

go →

4. Anticipated transfers

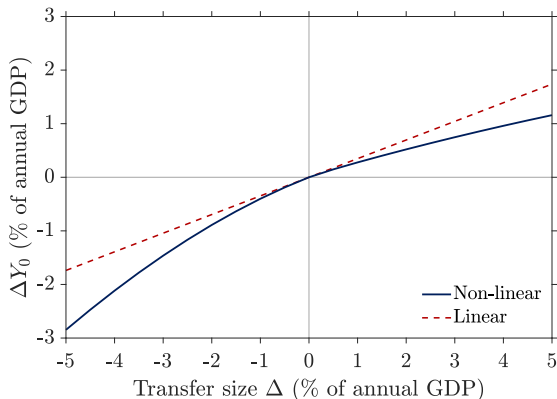
go →

5. Other shocks

go →

FISCAL CONTRACTIONS VS. EXPANSIONS

Output response

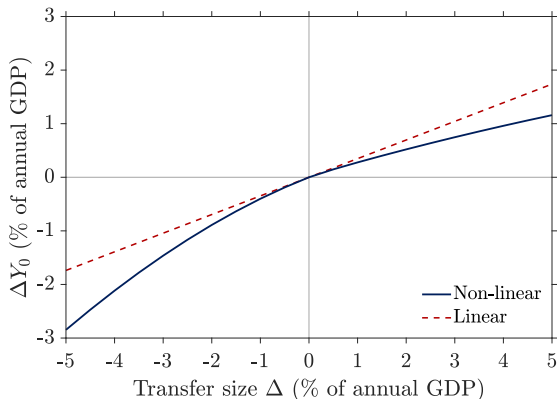


Concavity: linear **overstates** expansions, **understates** contractions.

Note: Baseline HANK model

FISCAL CONTRACTIONS VS. EXPANSIONS

Output response

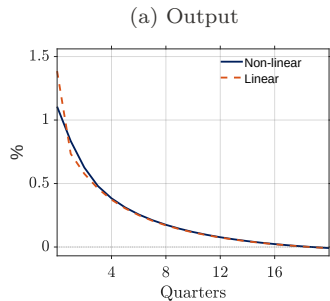


Concavity: linear **overstates** expansions, **understates** contractions.

Non-linearity slightly larger for contractions:

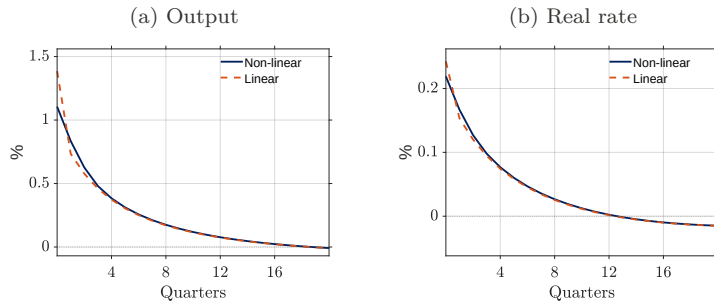
Note: Baseline HANK model

LINEAR METHOD UNDERSTATES PERSISTENCE



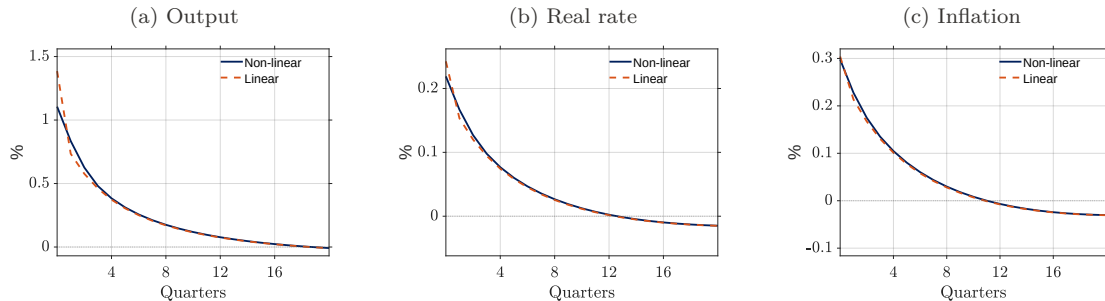
Note: Impulse responses in the HANK to a 1% of annual GDP transfer.

LINEAR METHOD UNDERSTATES PERSISTENCE



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LINEAR METHOD UNDERSTATES PERSISTENCE

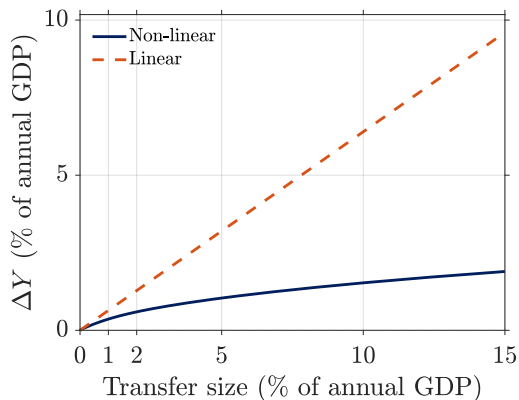


Note: Impulse responses in the HANK to a 1% of annual GDP transfer.

TARGETED TRANSFERS VS. UNTARGETED TRANSFERS

◀ OTHER RESULTS

(a) Output response (bottom 20%)

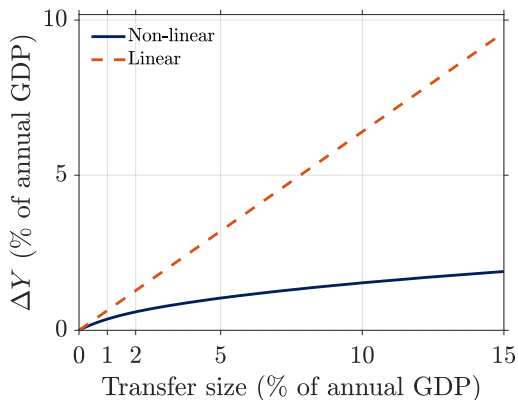


(b) Linear / non-linear

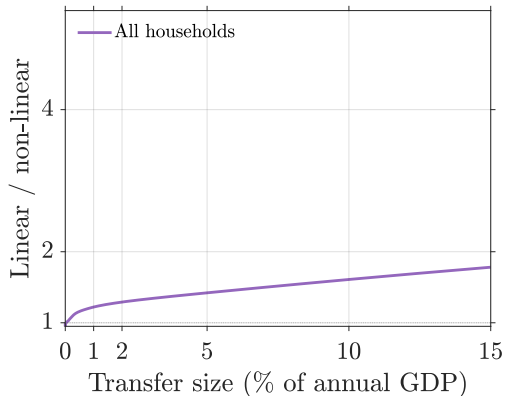
TARGETED TRANSFERS VS. UNTARGETED TRANSFERS

◀ OTHER RESULTS

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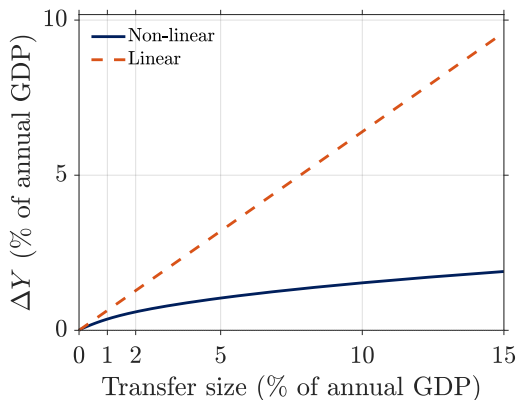
(b) Linear / non-linear



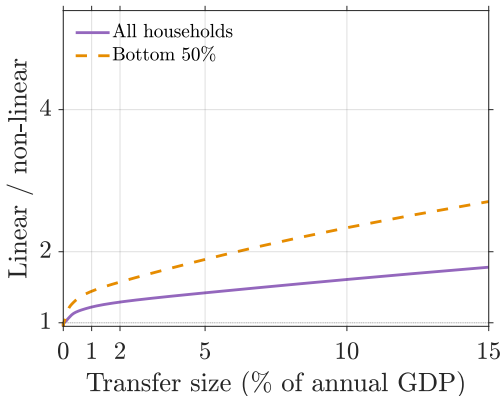
TARGETED TRANSFERS VS. UNTARGETED TRANSFERS

◀ OTHER RESULTS

(a) Output response (bottom 20%)



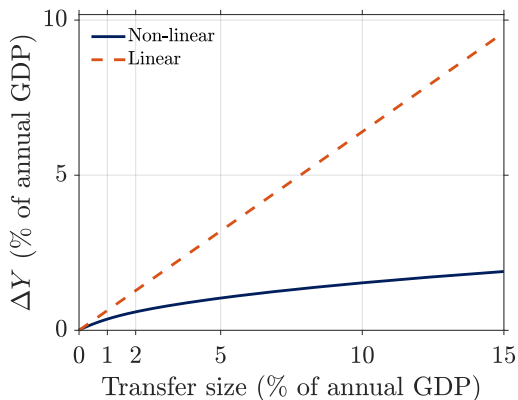
(b) Linear / non-linear



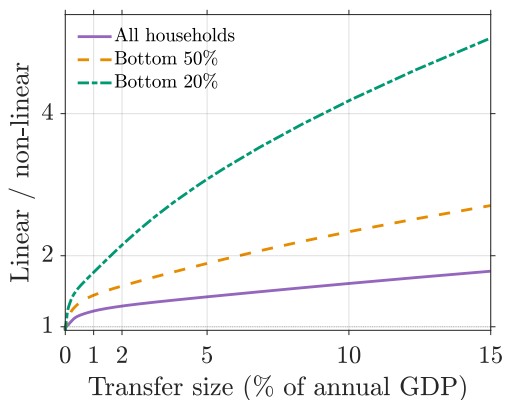
TARGETED TRANSFERS VS. UNTARGETED TRANSFERS

◀ OTHER RESULTS

(a) Output response (bottom 20%)



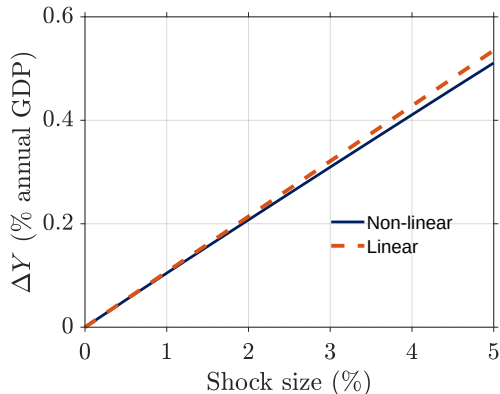
(b) Linear / non-linear



NON-LINEARITIES FOR MONETARY POLICY AND TFP

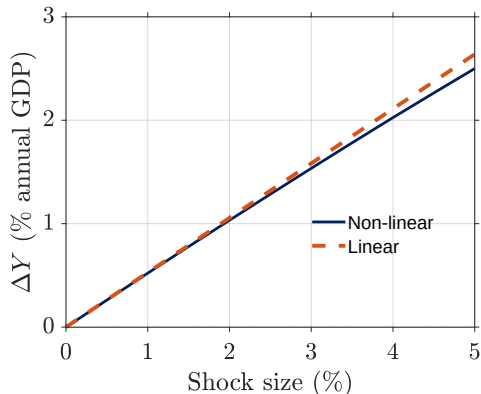
◀ OTHER RESULTS

(a) Monetary (rate cut)



► Tax-rate cut

(b) TFP



► Discount-factor (β) shock

CONCLUSIONS

- First-order methods significantly overstate the effects of fiscal transfers
 - ▶ Key: higher transfers shift households out of the borrowing constraint, so the aggregate MPC falls with transfer size

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 - ▶ Key: higher transfers shift households out of the borrowing constraint, so the aggregate MPC falls with transfer size
- For monetary policy, linear methods are accurate. But $HA \approx RA$
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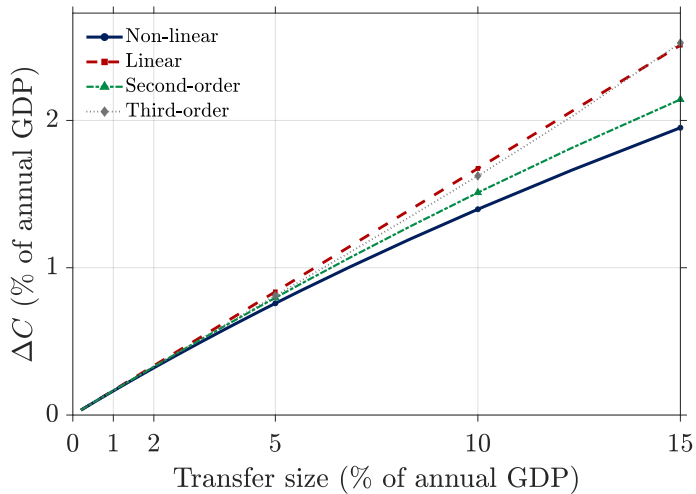
CONCLUSIONS

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 - ▶ Key: higher transfers shift households out of the borrowing constraint, so the aggregate MPC falls with transfer size
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 - ▶ Higher-order local approx. improves for narrow shocks, but deteriorate quickly

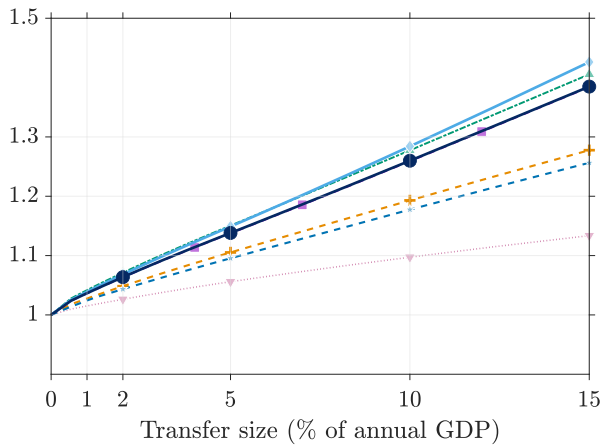
APPENDIX

Additional figures

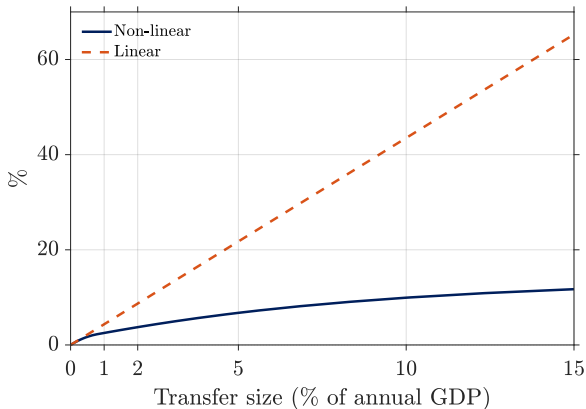
UNCONSTRAINED: HIGHER-ORDER LOCAL APPROX.



GE: FIRST-YEAR LINEAR-TO-NON-LINEAR RATIO



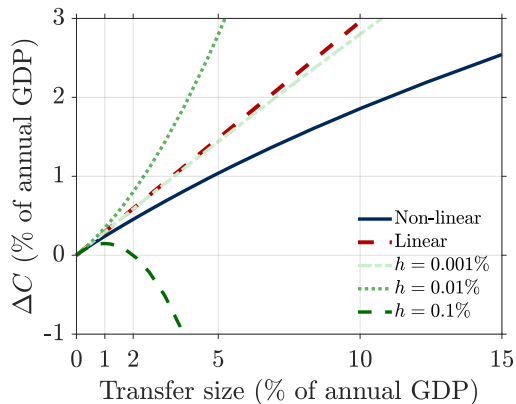
GE: FLEXIBLE PRICES WITH HOUSEHOLD LABOR CHOICE



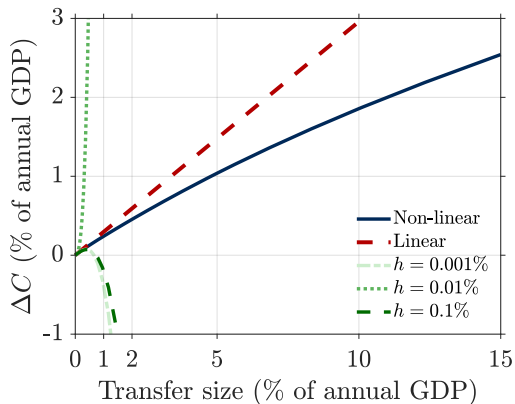
- Households choose labor freely (no union)
- Output effects via labor-supply incentives
- Real-rate response remains highly **concave**

HIGHER-ORDER WITH FINITE DIFFERENCES

(a) Second order



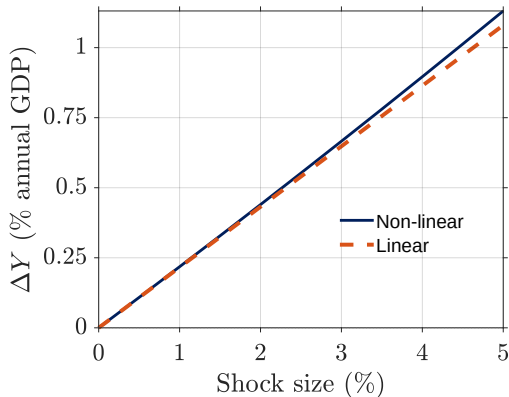
(b) Third order



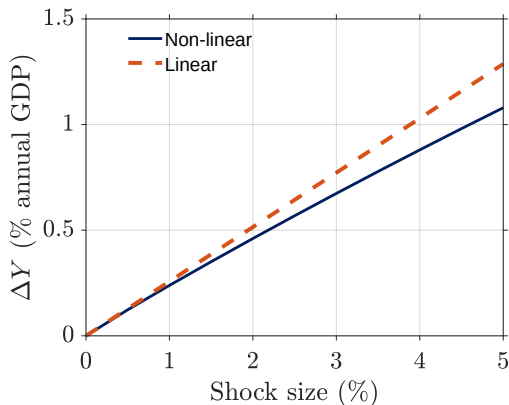
Extremely sensitive to step size and poorly behaved even for small shocks.

OTHER SHOCKS: DISCOUNT FACTOR AND LABOR TAX

(a) Discount-factor shock

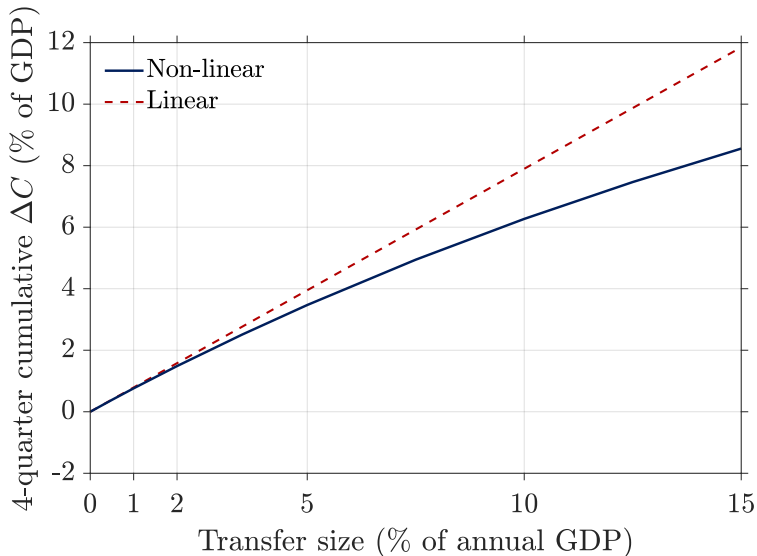


(b) Labor-tax cut

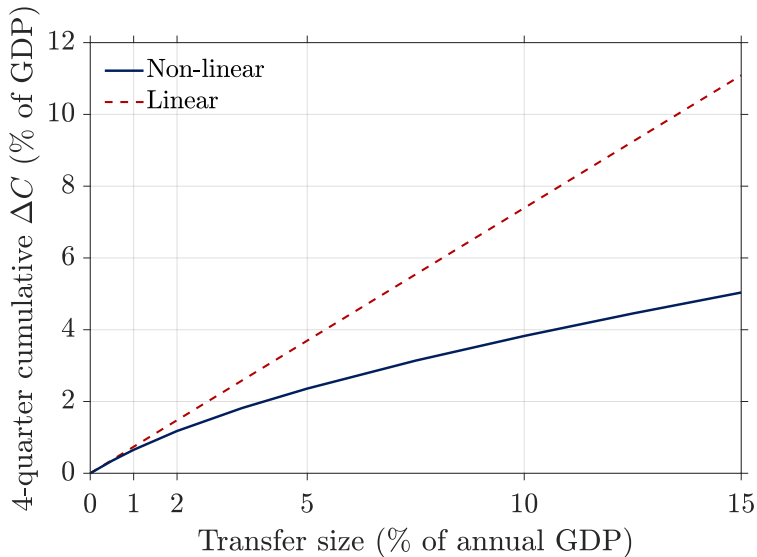


A τ cut shifts fewer resources to low-income, constrained households.

4-QUARTER CUMULATIVE RESPONSE (BASELINE HANK)

[← BACK](#)

4-QUARTER CUM. RESPONSE (BOTTOM-20% TARGETED)

[← BACK](#)

ALTERNATIVE DECOMPOSITION

[◀ BACK](#)

Exact non-linear **constrained** share + first-order Taylor for the **unconstrained**:

$$C_0^{\text{hyb}}(\Delta) = \int_Y \left[\underbrace{\int_{-\infty}^{\bar{x}(y)-\Delta} c(\tilde{x} + \Delta, y) dG(\tilde{x} | y)}_{\text{exact constrained share, MPC=1}} + \underbrace{\int_{\bar{x}(y)-\Delta}^{\infty} [c(\tilde{x}, y) + c_x^+(\tilde{x}, y) \Delta] dG(\tilde{x} | y)}_{\text{unconstrained: first-order Taylor}} \right] dF(y)$$

